

Relational Model - Library Media System (v3)

Page 1/4 | Lookup Tables | itemId = INT AUTO_INCREMENT

LOOKUP / ENUM TABLES

ItemType

Key	Column Name	Data Type	Nullable
PK	<u>type</u>	VARCHAR(50)	NOT NULL

MediaFormat

Key	Column Name	Data Type	Nullable
PK	<u>format</u>	VARCHAR(50)	NOT NULL

FanficType

Key	Column Name	Data Type	Nullable
PK	<u>type</u>	VARCHAR(50)	NOT NULL

SeriesInfo

Key	Column Name	Data Type	Nullable
PK	<u>seriesName</u>	VARCHAR(50)	NOT NULL

NEW LOOKUP TABLES

FilmType

Key	Column Name	Data Type	Nullable
PK	<u>type</u>	VARCHAR(50)	NOT NULL

BookFormat

Key	Column Name	Data Type	Nullable
PK	<u>bFormat</u>	VARCHAR(50)	NOT NULL

TranslationInfo

Key	Column Name	Data Type	Nullable
PK	<u>translationTypeOrLanguage</u>	VARCHAR(50)	NOT NULL

NEW: Genre

Key	Column Name	Data Type	Nullable
PK	<u>genre</u>	VARCHAR(50)	NOT NULL

NEW: Language

Key	Column Name	Data Type	Nullable
PK	<u>language</u>	VARCHAR(50)	NOT NULL

NEW: Fandom

Key	Column Name	Data Type	Nullable
PK	<u>fandom</u>	VARCHAR(100)	NOT NULL

CORE ENTITY

LibraryItem			
Key	Column Name	Data Type	Nullable
PK	itemId	INT AUTO_INCREMENT	NOT NULL
	title	VARCHAR(100)	NOT NULL
FK	itemType	VARCHAR(50)	NOT NULL
FK	genre	VARCHAR(50)	NOT NULL
FK	language	VARCHAR(50)	NOT NULL
FK	seriesName	VARCHAR(50)	NULL

FK: itemType -> ItemType(type)
FK: genre -> Genre(genre)
FK: language -> Language(language)
FK: seriesName -> SeriesInfo(seriesName)

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Page 3/4 | Subtype Tables (table-per-subclass, shared PK from LibraryItem)

SUBTYPE TABLES

Film

Key	Column Name	Data Type	Nullable
PK/FK	<u>itemId</u>	INT	NOT NULL
	director	VARCHAR(50)	NULL
FK	filmType	VARCHAR(50)	NOT NULL
FK	mediaFormat	VARCHAR(50)	NOT NULL
FK	translationInfo	VARCHAR(50)	NULL

FK: itemId -> LibraryItem(itemId)

FK: filmType -> FilmType(type)

FK: mediaFormat -> MediaFormat(format)

FK: translationInfo -> TranslationInfo(translationTypeOrLanguage)

Game

Key	Column Name	Data Type	Nullable
PK/FK	<u>itemId</u>	INT	NOT NULL
	creator	VARCHAR(50)	NULL

FK: itemId -> LibraryItem(itemId)

Book

Key	Column Name	Data Type	Nullable
PK/FK	<u>itemId</u>	INT	NOT NULL
FK	fandom	VARCHAR(100)	NULL
FK	fanficType	VARCHAR(50)	NULL
FK	bookFormat	VARCHAR(50)	NOT NULL

FK: itemId -> LibraryItem(itemId)

FK: fandom -> Fandom(fandom)

FK: fanficType -> FanficType(type)

FK: bookFormat -> BookFormat(bFormat)

TVSeries

Key	Column Name	Data Type	Nullable
PK/FK	<u>itemId</u>	INT	NOT NULL

FK: itemId -> LibraryItem(itemId)

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Page 4/4 | Junction Tables + Detail Tables

JUNCTION TABLES (multi-valued attributes)

Film_Actors

Key	Column Name	Data Type	Nullable
PK/FK	<u>filmId</u>	INT	NOT NULL
PK	<u>actorName</u>	VARCHAR(100)	NOT NULL

FK: filmId -> Film(itemId)

Book_Authors

Key	Column Name	Data Type	Nullable
PK/FK	<u>bookId</u>	INT	NOT NULL
PK	<u>authorName</u>	VARCHAR(100)	NOT NULL

FK: bookId -> Book(itemId)

DETAIL TABLES (TV Series structure)

Season

Key	Column Name	Data Type	Nullable
PK/FK	<u>tvSeriesId</u>	INT	NOT NULL
PK	<u>seasonNumber</u>	INT	NOT NULL

FK: tvSeriesId -> TVSeries(itemId)

Episode

Key	Column Name	Data Type	Nullable
PK/FK	<u>tvSeriesId</u>	INT	NOT NULL
PK/FK	<u>seasonNumber</u>	INT	NOT NULL
PK	<u>episodeNumber</u>	INT	NOT NULL
	episodeName	VARCHAR(100)	NULL

FK: (tvSeriesId, seasonNumber) -> Season(tvSeriesId, seasonNumber)

LEGEND

- PK** = Primary Key (underlined, identifies each row uniquely)
- FK** = Foreign Key (references a Primary Key in another table)
- PK/FK** = Both PK and FK (used in subtype tables - shared key from LibraryItem)
- NULL** = Optional column
- NOT NULL** = Required column

Green tables = NEW in v3